

CIP-50: Incentives, Capacity, and the Risk of a Low Leverage Parameter*

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Abstract

Although Cardano Improvement Proposal 50 (CIP-50, [Liesenfelt et al. \(n.d.\)](#)) comes in response to concerns raised by the current reward scheme, there is still limited systematic analysis of its consequences and implementation risks. In this paper, we complement existing analyses by evaluating both the proposal’s potential benefits and its main risks. On the benefit side, CIP-50 can strengthen pledge as a competitive margin, improving the relative position of well-pledged pools and reducing the attractiveness of large low-pledge pools. On the risk side, however, the effects depend critically on the leverage parameter. If set too low, the incumbent pool set may not provide enough reward-bearing capacity to accommodate all currently active stake under the current pledge distribution. Even when there is enough aggregate capacity, transition pressure may remain severe because spare capacity is fragmented and substantial stake reallocation may still be required. At the same time, recapitalization can soften that pressure, since the additional pledge needed to support current reward-relevant stake may be much smaller than the pure reallocation benchmark. The paper also highlights additional risks related to capital requirements, delegation instability, and strategic behavior. CIP-50 should therefore be evaluated not only for its potential incentive benefits, but also as a capacity and transition problem. We conclude that limiting these risks requires a relatively high leverage parameter at the cost of attenuating the proposal’s intended benefits.

Keywords: Cardano, CIP-50, proof of stake, delegation, staking rewards
JEL: D47, D61, G28, L14

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1 Introduction

Cardano Improvement Proposal 50 (CIP-50) is motivated by concerns that the current reward design does not make pledge a strong enough driver of decentralization. Under the existing scheme, stake size appears to matter more than pledge for delegators’ allocation decisions. As a result, large low-pledge pools may find it easier to attract stake, smaller well-pledged pools may struggle to compete, and multi-pool operators may face weaker constraints on Sybil-type strategies, to the detriment of decentralization. CIP-50 seeks to address this by strengthening pledge as a competitive margin with the aim of making large low-pledge pools less attractive and improving the position of smaller well-pledged pools.

In particular, CIP-50 seeks to strengthen that margin by tying each pool’s reward-bearing stake directly to its pledged amount. Under the proposal, pledge would constrain low-pledge pool rewards through an explicit cap on the stake that can earn rewards in each pool. (Liesenfelt et al., n.d.; Cardano Docs, n.d.; Brünjes et al., 2020; Morini, 2025)

CIP-50 therefore raises a broader question than whether it strengthens pledge in the intended way. It must also be assessed in terms of the implementation and incentive issues that may arise once reward-bearing capacity becomes pool-specific and pledge-dependent. This concern goes beyond marginal incentive effects. By making reward-bearing capacity depend more directly on pledge, CIP-50 turns the existing distribution of stake and pledge into a structural feature of the reform. This brings into focus several additional issues: whether the current pool set can accommodate all active stake in reward-bearing positions, how much transition pressure the reform may create through stake reallocation or additional pledge needs, and what coordination frictions, unintended incentives, or adjustment costs could arise during the transition.

In this paper, we study CIP-50 from that broader perspective. We first revisit the policy motivation for the proposal and clarify the competitive benefit it is intended to deliver under its design. We then develop a simple framework to evaluate, under the current pledge distribution, aggregate system capacity and the reallocation pressure implied by a given leverage choice, which yields a feasibility test for different values of L . Finally, we discuss implementation risks that are not captured by the feasibility test alone, including fragmented slack, stronger capital requirements, and delegation instability around pool-specific caps.

The central message is that the leverage parameter should be evaluated as a trade-off: it can strengthen pledge as a competitive margin, but it may also create capacity, transition, and implementation problems if set too low.

1.1 Summary of Findings

The main findings are as follows.

Summary of Findings

- **Weak role of pledge as incentive for delegation.** Under the current design, pledge appears too weakly connected to delegator-facing returns for most pools. In practice, stake size and fixed-cost dilution seem to matter more for competitive outcomes in the region where most pools operate.
- **Under CIP-50 pledge becomes a stronger competitive margin.** By tying reward-bearing capacity to pledge, CIP-50 can improve the relative position of well-pledged pools and reduce the attractiveness of large low-pledge pools.

- **Feasibility test.** For any value of the leverage parameter, the first policy question is whether the incumbent pool set provides enough total reward-bearing capacity. Low values of the leverage parameter can fail this test.
- **Feasibility does not eliminate transition pressure.** Passing the feasibility test does not guarantee a smooth transition. When the parameter is set close to the feasibility threshold, substantial amounts of stake may still need to move out of pools that already exceed their new quota.
- **Fragmented slack weakens feasibility.** Available capacity may be fragmented across many small pools, making stake reallocation harder in practice than the feasibility condition alone would suggest.
- **Added pledge can ease transition pressure.** The pledge gap, that is, the additional pledge required to retain existing stakes, is significantly narrower than the pressure for stake reallocation. Consequently, a modest increase in operator commitment may stabilize the transition without necessitating a massive migration of stakes.
- **Realized migration can fall short of accounting reallocation.** Some delegators may rationally remain in over-quota pools if those pools continue to offer relatively attractive net returns.
- **CIP-50 strengthens capital requirements.** A low leverage value can shift competitive advantage away from stake and toward access to capital. That is, CIP-50 may help some well-pledged smaller pools, but it can also raise entry barriers and favor operators with stronger access to capital.
- **Custodial-operator response.** CIP-50 can force large custodial operators to reconsider current reward-inefficient pool structures. Their response may range from more efficient pool structures, raising paid rewards and accelerating reserve depletion, to a reduction in participation.
- **Modest negative effect on reserve depletion.** CIP-50 may induce some large operators to adjust their pool structure, thereby increasing reserve depletion. However, the estimated magnitude of this effect appears to be modest.
- **Pool-specific caps create new vulnerability points.** Under the current design, the main reward threshold is the common saturation point. Under CIP-50, many pools would instead face their own pledge-based threshold, making pool attractiveness more sensitive to stake movements around those points and creating additional local vulnerability points.
- **Potential negative impact on UX.** Delegators may need to monitor both their own pool and the broader pool landscape more frequently to avoid these vulnerability points. This may worsen UX.

2 Motivation

To understand the motivation behind CIP-50, it is useful to begin with the current reward logic and the existing landscape of stake allocation and pool configurations.

2.1 Current reward logic

Under the current reward rule, the amount of stake that remains reward-bearing is capped at the common saturation point (z_0). Let k denote the target number of pools, T the total supply, and

$$z_0 = \frac{T}{k}$$

the saturation point.¹ For pool i , let $p_i > 0$ denote its declared (committed) pledge and $\sigma_i \geq p_i$ its total stake. Hence, reward-bearing stake is given

$$\tilde{\sigma}_i = \min\{\sigma_i, z_0\},$$

and reward-bearing pledge by

$$\tilde{p}_i = \min\{p_i, z_0\}.$$

Following [Cardano Docs \(n.d.\)](#), the gross reward of pool i can be written as

$$f(\sigma_i, p_i; q_i) = \frac{R}{1 + a_0} \left[\tilde{\sigma}_i + a_0 \tilde{p}_i \frac{\tilde{\sigma}_i - \tilde{p}_i \frac{z_0 - \tilde{\sigma}_i}{z_0}}{z_0} \right], \quad (1)$$

where R is the reward pot, a_0 is the pledge influence parameter, and q_i denotes the effective reward-bearing quota, which under the current design is common across pools ($q_i = z_0$ for every pool i).^{2,3,4}

The gross reward is maximized when $\sigma_i = p_i = z_0$ giving

$$f(z_0, z_0; z_0) = \frac{R}{k}.$$

while the case of a saturated pool with zero pledge ($\sigma_i = z_0, p_i = 0$) gives 23% less (when $a_0 = 0.3$):

$$f(z_0, 0; z_0) = \frac{R}{(1 + a_0)k}.$$

Thus, a purely self-funded pool, i.e. a pool with $\sigma_i = p_i$ needs a pledge $p_i \approx 82.9\% z_0$ to get the same gross rewards as a saturated pool with zero pledge:⁵

$$f(0.829z_0, 0.829z_0; z_0) = f(z_0, 0; z_0).$$

¹ $T \simeq 38.5B$ ADA as of March 2026 [Cardano ADA Supply Distribution, Epoch 616 \(2026\)](#)

²At the snapshot used in this paper (March 3, 2026, during epoch 616), the reward pot is approximately $R \simeq 15.6$ million ADA. This amount is obtained by taking 0.3% of reserves at that epoch, allocating 20% to the treasury and 80% to staking rewards. Strictly speaking, collected transaction fees should also be added, but we abstract from them because their contribution remains small relative to the reserve-funded component. A useful clarification is that the 15.6 million ADA figure would be distributed only if 100% of total supply were actively staked. Since only about 56% of total supply was staked at that time, the corresponding amount is about 8.5 million ADA. Finally, because rewards are normalized by $1 + a_0$, the amount effectively distributed is approximately 6.6 million ADA.

³ z_0, σ_i , and p_i enter in the formula in relative terms, as fractions of total supply T (e.g, this implies to use $1/k$ for the normalized z_0), while R is measured in ADA. Hence, the formula returns a fraction of the pot R that will be paid to pool i in the corresponding epoch.

⁴A pool with an active pledge below the declared pledge gets $f(\sigma_i, p_i; q_i) = 0$ in that epoch (see [Brünjes and Coutts \(2020\)](#))

⁵Assume, with slight abuse of notation, that p_i and z_0 are already normalized by total supply T , so that $z_0 = 1/k$. Then, using (1),

$$f(z_0, 0; z_0) = \frac{Rz_0}{1 + a_0}, \quad f(p_i, p_i; z_0) = \frac{R}{1 + a_0} \left[p_i + a_0 p_i \frac{p_i - p_i \frac{z_0 - p_i}{z_0}}{z_0} \right].$$

A more relevant comparison may be the ratio net reward of pool i per unit of stake, with the net reward given by the expression

$$f(\sigma_i, p_i; q_i) - c_i,$$

and where c_i is the fixed cost. Comparing

$$\frac{f(z_0, 0; z_0) - c_i}{z_0} \quad \text{vs.} \quad \frac{f(p_i, p_i; z_0) - c_i}{p_i}.$$

the ratio is equalized at $p_i \approx 31.8\%z_0$, which is still a high amount of capital.⁶ These capital thresholds should be read conditional on the current value of k , since $z_0 = T/k$ implies that the absolute ADA amount needed to reach any given fraction of saturation falls as k increases.⁷

The previous comparison of expression (1) helps to clarify why the current system is often perceived as only weakly pledge-sensitive for most active pools. Pledge does enter the exact reward formula, but for subsaturated pools its direct quantitative effect on member returns is typically modest relative to stake size and fees (see also (Morini, 2025)). This matters because most active pools are far below saturation.⁸

The protocol does not distribute $f(\sigma_i, p_i; q_i)$ directly as member returns: realized rewards are first adjusted by pool performance and then split according to fixed costs, margins, and stake shares. These additional layers determine how the limited direct effect of pledge on gross rewards is ultimately transmitted to the returns that delegators achieve.

In particular, Equation (1) is then adjusted for a pool's performance factor that we denote here with λ_i .⁹ Then, the realized gross reward is

$$\lambda_i f(\sigma_i, p_i; q_i).$$

Since $p_i - p_i \frac{z_0 - p_i}{z_0} = \frac{p_i^2}{z_0}$, the condition $f(z_0, 0; z_0) = f(p_i, p_i; z_0)$ becomes

$$1 = \frac{p_i}{z_0} + a_0 \frac{p_i^3}{z_0^3}.$$

Setting $a_0 = 0.3$ and solving gives $p_i \simeq 0.829z_0$.

⁶Proceeding as in the previous case, we obtain

$$\frac{R}{1 + a_0} \left[p_i + a_0 \frac{p_i^3}{z_0^2} \right] - c_i = \frac{p_i}{z_0} \left[\frac{Rz_0}{1 + a_0} - c_i \right].$$

Multiplying the left-hand side by $\frac{z_0}{z_0}$ and defining $x = \frac{p_i}{z_0}$, this can be rewritten as the cubic equation

$$\frac{R}{1 + a_0} a_0 z_0 x^3 + c_i x = c_i.$$

For the calibration, we set $a_0 = 0.3$, normalize $z_0 = 1/k = 1/500$, and take $c_i = 340$, a common fixed cost declared by pools. We also use $R \simeq 15.6$ million ADA. As reported by *Cardano ADA Supply Distribution, Epoch 616* (2026), reserves at epoch 616 were approximately 6.5 billion ADA, of which 0.3% is drawn each epoch and split between rewards (80%) and the treasury (20%).

⁷More broadly, k and CIP-50 can be seen as potentially complementary design margins: a higher k lowers the absolute pledge required for a given competitive position, while CIP-50 can make pool-splitting more pledge-constrained by requiring pledge support in each pool.

⁸In the March 3, 2026 snapshot used in this paper, the median implied saturation point is approximately 76.940 million ADA, yet only 7 active pools are currently at or above full saturation. By contrast, there are 2,717 active pools with total active stake of 21.569 B ADA and total active pledge of 2.932 B ADA. The current design therefore operates predominantly in a region where the effective reward-bearing quota at z_0 is not binding.

⁹The performance adjustment factor is of the form $\lambda_i = \beta/\sigma_a$, where β is the fraction of all blocks actually produced by the pool during the epoch and σ_a is the pool's stake relative to total active stake. In comparative statics or policy exercises it is often convenient to set $\lambda_i = 1$ in order to isolate the incentive effects of stake, pledge, and fees.

Assume $\lambda_i = 1$, let $c_i \geq 0$ denote the fixed cost charged by the pool and $m_i \in [0, 1)$ its margin. For pool i , the protocol first pays the fixed cost c_i whenever $f(\sigma_i, p_i; q_i) > c_i$. The remaining amount, $[f(\sigma_i, p_i; q_i) - c_i]_+$, is then allocated as follows: a fraction m_i is taken by the operator as the pool margin, that is, as a commission on delegation rewards, and the residual fraction $(1 - m_i)$ is distributed proportionally among all stake delegated to the pool, including the operator’s own pledged stake. Thus, the pool operator gets:

$$\begin{cases} c_i + (f(\sigma_i, p_i; q_i) - c_i)[m_i + (1 - m_i)\frac{\hat{p}_i}{\sigma_i}], & \text{if } f(\sigma_i, p_i; q_i) > c_i, \\ f(\sigma_i, p_i; q_i), & \text{otherwise} \end{cases},$$

where $\hat{p}_i$ denotes the operator’s active pledge, and a delegator d with stakes σ_d

$$\begin{cases} (1 - m_i)(f(\sigma_i, p_i; q_i) - c_i)\frac{\sigma_d}{\sigma_i}, & \text{if } f(\sigma_i, p_i; q_i) > c_i, \\ 0, & \text{otherwise} \end{cases}.$$

The expressions above inform how the current design rewards stake and pledge. A natural next question is how the market (pools and delegators) responds to that design. We next present data as of March 3, 2026, which provides a snapshot of the current staking landscape.

2.2 Data snapshot

The following table summarizes a data snapshot as of March 3, 2026 from [Cexplorer](#) and [Cardano ADA Supply Distribution, Epoch 616 \(2026\)](#) websites, and it is based only on pools with positive active stake. Appendix A provides additional descriptive statistics.

Metric	Value
Active pools	2,717
Total supply	38.47 B ADA
Total active stake	21.57 B ADA
Total active pledge	2.93 B ADA
Median implied z_0	76.94 M ADA
Saturated pools ($\sigma_i \geq z_0$)	7
Zero-pledge active pools	225
Stake in zero-pledge active pools	3.58 B ADA

Table 1: Core statistics. Rounded numbers. Active-pool snapshot of March 3, 2026, during epoch 616. Zero-pledge active pools figure only considers pools with pledge strictly 0 ADA

Two facts in Table 1 are central. First, the current saturation point binds for very few pools. Second, the table also shows that a nontrivial amount of active stake remains allocated to zero-pledge pools. This is only part of the picture, however. Beyond these zero-pledge pools, there are many additional pools operating with low pledge, as the histogram below illustrates.

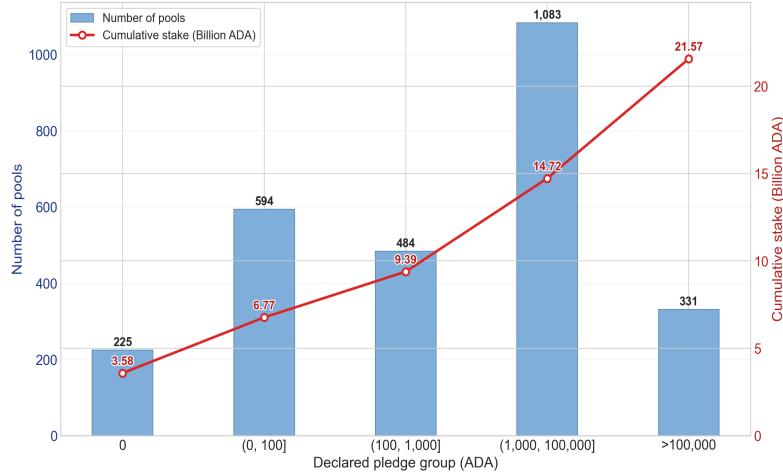


Figure 1: Number of pools by declared pledge group and cumulative stake

The histogram sharpens the point that the nontrivial stake allocated to zero-pledge pools is not an isolated feature of the data. Many additional pools also operate with very low pledge, and together these low-pledge groups account for a substantial share of active stake.¹⁰

2.3 Observed patterns of the current design

Taken together, Table 1 and Figure 1 show that a substantial amount of delegation remains allocated to pools with zero or very low pledge, and suggest that the current design operates mostly in a region where the saturation point rarely binds.

The empirical exercise in Appendix B provides additional support for this view. Looking at how pool characteristics are associated with subsequent changes in delegation, the data reveal a clear pattern: pools with more stake tend to attract more delegator growth, whereas higher pledge does not display a similarly clear relationship.

Overall, the evidence suggests that, under the current system, scale matters more than pledge in shaping delegation outcomes. As a result, small but well-pledged pools may struggle to compete with larger pools that already control substantial stake. This asymmetry reflects not only the limited role of pledge in the current reward scheme, but also other forces that favor scale. In particular, fixed operating costs are easier to dilute in larger pools, which improves delegator returns, and larger pools also tend to deliver smoother realized rewards because block production is less volatile in the short term at higher stake.¹¹

The competitive advantage of large operators can be extended through Sybil-type strategies, that is, by creating additional low-pledge pools that still receive meaningful delegation. Smaller pools, by contrast, cannot rely on scale and therefore need pledge to compete. The result is stake that is increasingly centralized among a limited number of large multi-pool operators. In line with this concern, Liesenfelt et al. (n.d.) claim, as of epoch 559, that just 25 operators control more than 50% of total stake, and that many of them operate multiple pools, often with low pledge.

This pattern helps explain why CIP-50 was proposed. If pools with little or no pledge can still attract substantial stake and earn meaningful rewards, then “skin in the game” is not playing a

¹⁰See Appendix D to see a list of zero-declared pledge and (0, 1000]-declared pledge main operators.

¹¹One way to mitigate the advantage that fixed-cost dilution gives to larger pools is to reduce the minimum fixed cost that pools are allowed to declare. This raises net rewards for all pools declaring the new lower cost, but the increase in reward per unit of stake is greater for smaller pools, thereby partially offsetting the scale advantage of larger ones.

strong disciplining role in the current allocation.

Finding: Limited role of pledge in delegation decisions

Under the current design, pledge appears too weakly connected to delegator-facing returns for most pools. In practice, stake size and fixed-cost dilution seem to matter more for competitive outcomes in the region where most pools operate.

2.4 What CIP-50 aims to change

The evidence points to several undesirable and connected patterns:

- Stake carries more weight than pledge in reward outcomes and appears to matter more in shaping delegators' allocation decisions.
- This characteristic may explain the presence of Sybil-type behavior, particularly among large operators that control multiple pools.
- As a result, small but well-pledged pools may find it difficult to compete with larger pools that already command substantial stake.
- In turn, these dynamics may contribute to the centralization of stake among a relatively small number of large operators.

These patterns also help explain the logic of CIP-50. The proposal reflects the view that the current parameters k and a_0 have not been sufficient either to achieve a satisfactory degree of decentralization or to deter Sybil behavior strongly. It therefore introduces a new parameter that ties reward-bearing capacity more directly to pledge by limiting how much reward-bearing stake a pool can support. The intended effect is to strengthen pledge as a competitive margin, thereby making large low-pledge pools less attractive, increasing the cost of multi-pool Sybil strategies, improving the position of smaller but well-pledged pools, and ultimately mitigating centralization.

More formally, CIP-50 changes the pool-level reward-bearing quota from the current one $q_i = z_0$ to

$$q_i(L) = \min\{z_0, p_i L\},$$

where L is the pledge leverage parameter giving a pool specific cap $p_i L$. Accordingly, reward-bearing stake becomes

$$\tilde{\sigma}_i = \min\{\sigma_i, z_0, p_i L\} = \min\{\sigma_i, q_i(L)\},$$

and the corresponding gross reward is

$$f(\sigma_i, p_i; q_i(L)).$$

Importantly, this does *not* impose a hard quota on how much stake a pool can receive. Pools can still attract stake above $q_i(L)$. The change is narrower and more consequential: stake above $q_i(L)$ no longer remains reward-bearing. In other words, CIP-50 is a quota on reward-bearing capacity, not on delegation itself.

Seen this way, CIP-50 does not alter the general structure of the reward formula. What it changes is the amount of stake that enters it as reward-bearing. Under the current design, all stake up to the common saturation point z_0 is reward-bearing. Under CIP-50, pools with sufficiently high pledge continue to operate under that same limit, but pools with low pledge face a tighter pool-specific cap $p_i L$. This is the mechanism through which the proposal seeks to strengthen the role of pledge in staking competition.

Consider the gross reward at equation (1) when $q_i(L) = \min\{z_0, p_i L\}$. It is still maximized when $\sigma_i = p_i = z_0$ for any $L > 1$

$$f(z_0, z_0; q_i(L)) = \frac{R}{k}.$$

while the case of a saturated pool with zero pledge (implying now $q_i(L) = 0$) gets:

$$f(z_0, 0; 0) = 0.$$

These two cases describe the extremes of the schedule under CIP-50. To understand the mechanism more fully, it is also useful to consider intermediate pledge levels for pools with total stake fixed at $\sigma_i = z_0$. The intended logic of CIP-50 is to improve the relative position of well-pledged small and medium pools by making pledge a stronger competitive margin. The counterpart is that pools with low pledge become more constrained. Figure 2 illustrates this effect: for intermediate pledge levels, plotted as p_i/z_0 (in percent and truncated at 30% for visibility), low-pledge pools can be penalized sharply when L is set too low.

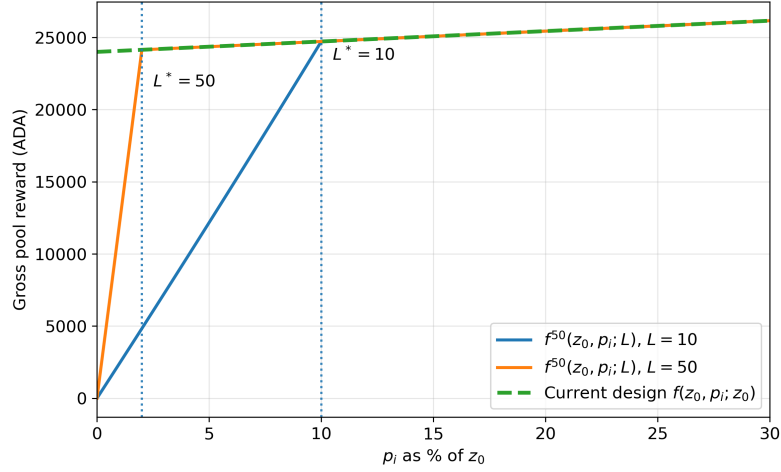


Figure 2: Pool rewards as a function of pledge under CIP-50. The figure considers pools with identical stake, $\sigma_i = z_0 = T/k$. High-pledge pools earn more, while low-pledge pools are penalized; when L is low, this penalty applies even at moderate pledge levels. Parameters: $R = 15.6\text{M ADA}$, $a_0 = 0.3$, and $k = 500$.

We can also see the effect of CIP-50 by comparing net rewards per unit of stake after fixed costs. Figure 3 illustrates

$$\frac{\max\{f(\sigma_i, p_i; q_i(L)) - c_i, 0\}}{\sigma_i}$$

under the current design and under CIP-50 for different pledge levels. Once fixed costs are taken into account, the competitive effect of the reform becomes clearer. A pool must be large enough to dilute the fixed cost, but under CIP-50 it must also have enough pledge to avoid losing reward-bearing capacity too early.

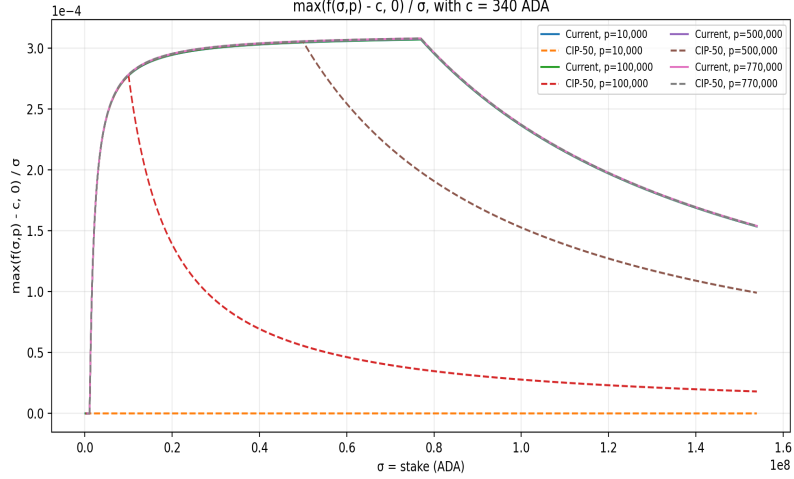


Figure 3: Illustrative net reward per unit of stake after fixed cost under the current design and under CIP-50. Parameters: $z_0 = 77M$ ADA, $p \in \{10\text{ K}, 100\text{ K}, 500\text{ K}, 770\text{ K}\}$, $L = 100$, $R = 15.6M$ ADA, $T = 38.5B$ ADA, $a_0 = 0.3$, and fixed cost $c = 340$ ADA.

Figure 3 makes this trade-off visible. For low-pledge pools, CIP-50 sharply compresses the stake range over which net returns per unit of stake remain attractive, as the pool reaches its cap $p_i L$ relatively early. By contrast, sufficiently well-pledged pools are much less affected, so their competitive position after fixed costs remains close to that under the current design.

Finding: Pledge becomes a stronger competitive margin

By tying reward-bearing capacity to pledge, CIP-50 can improve the relative position of well-pledged pools and reduce the attractiveness of large low-pledge pools.

Hence, the pool-level differentiation introduced by CIP-50 constrains low-pledge pools and can improve the relative position of better-pledged ones. But the reform also raises broader questions. Once reward-bearing capacity becomes pledge-dependent, can the existing pool set still accommodate all active stake? Would limited capacity or migration frictions create stress during the transition? How would delegators respond, and what new incentive vulnerabilities might emerge? The next sections examine these issues.

3 A Unified Model of Reallocation Under CIP-50

The previous section explained the pool-level logic of CIP-50. We now turn to its system-level implications: whether the incumbent pool set can still accommodate all currently active stake once reward-bearing capacity becomes pledge-dependent, and how much reallocation this may require.

3.1 The feasibility problem

The analysis in this section keeps the incumbent pool set and the observed pledge distribution fixed at their snapshot values. It should therefore be read as a mechanical benchmark for feasibility and transition pressure under the current configuration: adjustment occurs only through stake reallocation, abstracting from additional pledge, recapitalization, or internal reallocation of pledge within operator groups. It is not a full equilibrium analysis with endogenous pledge adjustment. We begin from this fixed-pledge benchmark and then contrast it with the opposite extreme of pure recapitalization.

3.1.1 Setup

Let $P = \{1, \dots, n\}$ denote the set of active pools. Pool i has current stake $\sigma_i \geq 0$ and pledge $p_i \geq 0$. Let's denote total stake as

$$S = \sum_{i \in P} \sigma_i.$$

As already stated, under CIP-50 with leverage parameter $L > 0$, the maximum amount of stake that can remain reward-bearing in pool i is

$$q_i(L) = \min\{z_0, p_i L\}.$$

Thus, we define the aggregate reward-bearing quota by

$$Q(L) = \sum_{i \in P} q_i(L).$$

To study reallocation, we deliberately use a minimal model. A post-reform allocation is a vector $x(L) = (x_i(L))_{i \in P}$ satisfying $x_i(L) \geq 0$ for each pool. Thus, a *fully reward-bearing allocation* under L is any allocation such that

$$x_i(L) \leq q_i(L) \quad \text{for every } i \in P,$$

and

$$\sum_{i \in P} x_i(L) = S.$$

In words, all current stake remains staked and all of it fits inside reward-bearing pool quotas.

This setup strips away preference heterogeneity, search frictions, and strategic timing in order to isolate the first-order feasibility question. If a fully reward-bearing allocation does not exist even in this reduced environment, then a low L fails before one introduces any additional friction.

3.1.2 A feasibility criterion

Proposition 1. *The function $Q(L)$ is weakly increasing, continuous, and concave in L .*

Proof. Each term $\min\{z_0, p_i L\}$ is weakly increasing, continuous, and concave in L . Summing over pools preserves these properties. $\square$

Proposition 2 (Reward-Bearing Feasibility Criterion). *For any fixed leverage parameter $L > 0$, a fully reward-bearing allocation exists if and only if*

$$Q(L) \geq S.$$

Proof. If a fully reward-bearing allocation exists, then

$$S = \sum_{i \in P} x_i(L) \leq \sum_{i \in P} q_i(L) = Q(L).$$

Conversely, if $Q(L) \geq S$, choose any sequence of allocations filling pools up to $q_i(L)$ and until all S stake has been assigned. Because aggregate quota is at least S , this construction terminates with an allocation satisfying $x_i(L) \leq q_i(L)$ for all i and $\sum_i x_i(L) = S$. $\square$

Corollary 1. *If $Q(L) < S$, then at least*

$$S - Q(L)$$

stake cannot be placed into reward-bearing positions after CIP-50 is introduced.

Proof. The total volume of reward-bearing positions across all pools is exactly $Q(L)$. If $S > Q(L)$, the residual amount $S - Q(L)$ cannot fit inside those positions. $\square$

Definition 1. Define the critical leverage threshold by

$$L^* = \inf\{L > 0 : Q(L) \geq S\}.$$

The criterion in Proposition 2 is algebraically simple. Its value is not mathematical sophistication but policy clarity: once CIP-50 changes the definition of reward-bearing stake, the accounting identity $Q(L) \geq S$ becomes the first admissibility test for any proposed leverage value.

Proposition 2 says that L^* is the smallest leverage value at which the incumbent pool set can still host all currently active stake in reward-bearing positions.

3.1.3 Transition pressure from the current allocation

The feasibility criterion gives an existence condition, but existence alone is not enough for policy. The current stake vector matters because CIP-50 would be introduced into an already populated system.

Define the currently over-quota stake at leverage L by

$$E(L) = \sum_{i \in P} \max\{\sigma_i - q_i(L), 0\}.$$

This is the amount of stake that, under the current allocation, already sits above the CIP-50 reward-bearing quota. In that sense, $E(L)$ can also be read as a stress-test object: it measures the transition pressure that would arise if the system had to adjust through stake reallocation alone, without offsetting pledge increases.

Proposition 3. Fix $L > 0$. Suppose the current stake vector is $\sigma = (\sigma_i)_{i \in P}$ and suppose $x(L)$ is any fully reward-bearing allocation under L . Then at least $E(L)$ stake must be moved away from pools that are currently above their quota.

Proof. For every pool with $\sigma_i > q_i(L)$, any fully reward-bearing allocation must satisfy

$$x_i(L) \leq q_i(L) < \sigma_i.$$

Hence pool i must lose at least $\sigma_i - q_i(L)$ stake. Summing over all currently over-quota pools gives a required outflow of at least

$$\sum_{i \in P} \max\{\sigma_i - q_i(L), 0\} = E(L).$$

$\square$

Proposition 3 is important because it measures transition pressure, not just feasibility. Even when $Q(L) \geq S$, the system may still require very large stake movement to reach a fully reward-bearing post-reform allocation.

3.1.4 Pledge adjustment under the current allocation

$E(L)$ measures adjustment purely through *stake reallocation*. Another extreme would be an adjustment purely through *recapitalization*, i.e., through newly added pledge.

Define $G(L)$ as the pledge gap, i.e., the minimum additional declared pledge that would be needed to support the current reward-relevant stake profile at leverage L . Formally,

$$G(L) := \sum_{i \in P} \max \left\{ \frac{\sigma_i^*}{L} - p_i, 0 \right\}, \quad \sigma_i^* := \min \{ \sigma_i, z_0 \}.$$

Proposition 4. *Fix $L > 0$ and suppose the current stake vector is $\sigma = (\sigma_i)_{i \in P}$, with $\sigma_i^* := \min \{ \sigma_i, z_0 \}$. Let the current declared pledge vector be $p = (p_i)_{i \in P}$. Suppose the current stake allocation is kept fixed, and consider the problem of choosing a new declared pledge vector $p' = (p'_i)_{i \in P}$ such that every pool can support its current reward-relevant stake under CIP-50, that is, $\sigma_i^* \leq \min \{ z_0, p'_i L \}$ for every $i \in P$. Then the minimum total additional pledge required is*

$$G(L) := \sum_{i \in P} \max \left\{ \frac{\sigma_i^*}{L} - p_i, 0 \right\}.$$

Proof. First, we show that any feasible p' must satisfy total added pledge larger or equal to $G(L)$. Then, we show that there exists a feasible p' that attains $G(L)$.

Fix any pool $i \in P$. Since $\sigma_i^* \leq z_0$ by definition, the constraint $\sigma_i^* \leq \min \{ z_0, p'_i L \}$ implies $\sigma_i^* \leq p'_i L$. Hence, for pool i , any feasible new pledge must satisfy

$$p'_i \geq \max \left\{ p_i, \frac{\sigma_i^*}{L} \right\}.$$

It follows that the additional pledge at pool i must be at least

$$p'_i - p_i \geq \max \left\{ \frac{\sigma_i^*}{L} - p_i, 0 \right\}.$$

Summing across pools gives that any feasible pledge adjustment must satisfy

$$\sum_{i \in P} (p'_i - p_i) \geq \sum_{i \in P} \max \left\{ \frac{\sigma_i^*}{L} - p_i, 0 \right\} = G(L),$$

indicating that $G(L)$ is a lower bound on the total additional pledge required.

To show that this bound is attained, define

$$p'_i := \max \left\{ p_i, \frac{\sigma_i^*}{L} \right\} \quad \text{for each } i \in P.$$

Then, for every pool i , $p'_i L \geq \sigma_i^*$, so $\sigma_i^* \leq \min \{ z_0, p'_i L \}$, p' is feasible. Since

$$p'_i - p_i = \max \left\{ \frac{\sigma_i^*}{L} - p_i, 0 \right\},$$

it follows that

$$\sum_{i \in P} (p'_i - p_i) = \sum_{i \in P} \max \left\{ \frac{\sigma_i^*}{L} - p_i, 0 \right\} = G(L),$$

and the minimum total additional pledge required is exactly $G(L)$. $\square$

Proposition 4 complements Proposition 3 by identifying a distinct adjustment margin. While $E(L)$ measures the minimum stake that must move under a pure reallocation benchmark, $G(L)$ measures the minimum additional pledge required under a pure recapitalization benchmark. The transition problem created by CIP-50 is inherently two-dimensional: it may require stake to move, pledge to be added, or some combination of both. Hence, a value of L that looks difficult under the fixed-pledge benchmark may become less problematic if operators can add pledge.

3.1.5 Empirical Calibration

The previous subsection established the relevant accounting objects. We now quantify them in the March 3, 2026 snapshot.

Using the active-pool snapshot, we compute $Q(L)$ and $E(L)$ for each integer value of L from 1 to 500. We find that the feasibility threshold lies between $L = 140$ and $L = 141$. Hence the smallest admissible integer value in the current data is $L = 141$.

L	Pools above $q_i(L)$ count	Over-quota stake $E(L)$ [B ADA]	Reward-bearing slack $Q(L) - S$ [B ADA]	Forced shortfall $\max\{S - Q(L), 0\}$ [B ADA]	Reward-bearing free space [B ADA]	Pledge Gap $G(L)$ [B ADA]	$G(L)/E(L)$ %
100	1045	14.641	-3.821	3.821	10.819	0.145	0.99
117	1022	14.290	-2.179	2.179	12.111	0.121	0.84
140	984	13.852	-0.055	0.055	13.798	0.098	0.71
141	983	13.834	0.036	0.000	13.870	0.097	0.70
153	969	13.631	1.121	0.000	14.752	0.088	0.65
200	906	13.003	4.652	0.000	17.656	0.064	0.49
250	870	12.533	8.095	0.000	20.628	0.049	0.39
300	837	12.153	11.263	0.000	23.417	0.040	0.33
350	813	11.850	14.028	0.000	25.877	0.033	0.28

Table 2: Selected leverage values using the current sweep in the repository. Over-quota stake is $E(L) = \sum_i \max\{\sigma_i - q_i(L), 0\}$. Reward-bearing free space is $\sum_i \max\{q_i(L) - \sigma_i, 0\}$.

Table 2 sharpens the policy discussion.

- At $L = 100$, the system fails the basic feasibility test: $Q(100) - S \approx -3.821$ B ADA. By Corollary 1, at least 3.821 B ADA cannot remain in reward-bearing positions.
- At $L = 141$, aggregate feasibility is restored, but only barely. The system has approximately 0.036 B ADA of aggregate slack, which is around 36 million ADA against a total active stake of 21.569 B ADA.
- Even at $L = 141$, Proposition 3 implies that at least 13.834 B ADA must move away from pools that are currently above quota to reach a fully reward-bearing allocation.
- Higher values such as 153 or 200 provide larger aggregate buffers, but they still leave substantial reallocation pressure.
- The pure-recapitalization benchmark ($G(L)$) is modest relative to ($E(L)$): for example, at ($L = 141$) total additional declared pledge on the order of 0.1B ADA would be enough to cover current reward-relevant stake if stake did not migrate.

An important next question, which the present calibration does not resolve, is how concentrated this pledge gap is across operators. If much of it is concentrated in a small number of large custodial or enterprise actors, the practical adjustment problem may be narrower than the fixed-pledge benchmark suggests.

Table 2 highlights selected leverage values; Figure 4 shows the same trade-off over the full range of L .

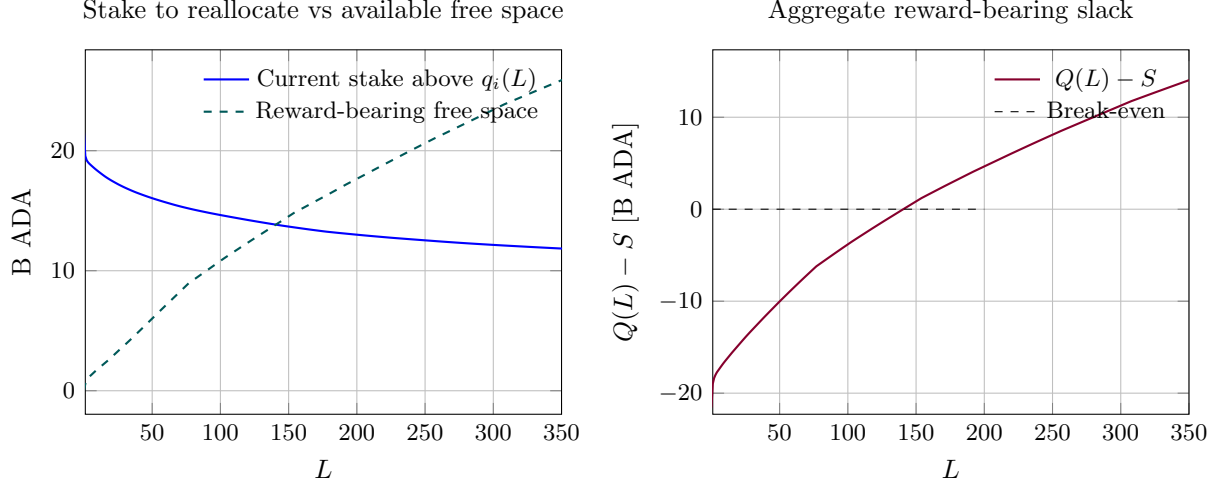


Figure 4: The left panel compares the amount of stake currently above the post-CIP-50 reward-bearing quota with the amount of reward-bearing slack capacity available elsewhere. The right panel gives the aggregate feasibility margin $Q(L) - S$. For policy, the left panel is the transition problem and the right panel is the feasibility test.

Figure 4 is the empirical core of the paper. The right panel identifies the feasibility threshold. The left panel shows the transition problem directly: how much stake currently needs to be accommodated elsewhere, and how much reward-bearing room exists to receive it. Threshold-adjacent values remain operationally uncomfortable because the stake requiring relocation remains very large even once aggregate feasibility turns positive.

Even when aggregate room exists, the practical use of that room may be limited by how it is distributed across pools. Next, we make that point explicit before turning to delegator behavior.

3.1.6 Fragmented slack is not the same as usable slack

Another important implementation risk is fragmented slack. The feasibility criterion asks whether the system has enough reward-bearing capacity in the aggregate, while the transition-pressure measure indicates how much stake would need to be reallocated. But even when the feasibility condition is satisfied, additional frictions may still arise. Delegators do not move stake into “the aggregate.” They must identify specific pools with remaining room, assess whether those pools can absorb their stake and still offer acceptable rewards, and reallocate at the right time.

This distinction matters quantitatively. At $L = 141$, aggregate reward-bearing free space is about 13.870 B ADA. But that free space is distributed across 1734 pools, while 983 pools are above the CIP-50 quota. The average free-space pocket is therefore only about 8.0 million ADA.

Why is this a policy risk? Because decentralized reallocation requires more than enough reward-bearing capacity in the aggregate:

- delegators must be able to identify pools that still have reward-bearing room;
- those pools must also remain attractive once fees, reputation, expected variance, and other pool characteristics are taken into account;

- available room may disappear quickly as many delegators compete for the same limited slack pockets;
- if many pools operate close to their pledge-based quota, reallocation may require repeated monitoring and further adjustments.

We do not claim that fragmented slack will imply exit. But the policy implication is still strong: values of L that barely satisfy $Q(L) \geq S$ should not be interpreted as safe, because aggregate capacity may be practically difficult to use.

We now summarize the main findings of this subsection.

Finding: Hard feasibility under the current pledge distribution

If $L < L^*$, the system does not have enough reward-bearing capacity to accommodate all currently active stake under the current pledge distribution.

Finding: Transition pressure

Even when L is just above L^* , the required reallocation can still be very large. $E(L)$ measures how much of stakes would still need to move if all active stake were to remain in reward-bearing positions.

Finding: Fragmented slack weakens feasibility

Even when the system has enough aggregate capacity, that capacity may be distributed across many small pools. As a result, threshold-adjacent values of L can remain operationally fragile because reallocating stake is harder than the aggregate capacity measure suggests.

Finding: Recapitalization can ease transition pressure

The pledge gap $G(L)$ is much smaller than the stake-reallocation measure $E(L)$. This suggests that, for some operators, relatively modest additional pledge could relax transition pressure without requiring equally large stake migration.

These findings identify a feasibility threshold and a distinct reallocation problem. What they do not show is the exact behavioral path of delegators, the speed of migration, or the amount of temporary exit from staking. Even more, whether that pressure translates into realized migration is a separate behavioral question, since delegators respond to expected net returns rather than to feasibility conditions alone. We discuss this next.

3.2 Delegator incentives and realized migration

The feasibility analysis above identifies how much stake would need to move in a fully reward-bearing allocation. The next question is whether delegators will in fact move in that amount.

The answer is not immediate, because delegators compare expected net returns across pools rather than feasibility conditions. Under CIP-50, once a pool moves above its effective quota $q_i(L) = \min\{z_0, p_i L\}$, its reward per unit of stake begins to deteriorate. But deterioration does not necessarily imply immediate outflows. A pool may remain attractive if its net return is still higher than that of smaller alternatives, for example because fixed costs are better diluted. Example 1 illustrates this point with two pools.

Example 1. Assume $k = 500$, $L = 100$, $a_0 = 0.3$, a total supply of $T = 38.5B$ ADA, $z_0 = T/k = 77M$ ADA, and $R = 15.6M$. Consider two pools with the same fixed cost $c = 340$ ADA. Pool A is large but weakly pledged, while Pool B is smaller but better pledged:

	Pool A	Pool B
Stake σ_i	30 M ADA	2 M ADA
Pledge p_i	0.20 M ADA	0.60 M ADA
$p_i L$	20 M ADA	60 M ADA
Quota reached	Yes	No

At these values, Pool A is already above its quota, while Pool B is not. Yet net reward per unit of stake remains higher in Pool A:¹²

$$\frac{f(\sigma_A, p_A; p_A L) - c}{\sigma_A} \simeq \frac{3150.7}{30M} \simeq 0.0196\%,$$

while the ones for Pool B are:

$$\frac{f(\sigma_B, p_B; p_B L) - c}{\sigma_B} \simeq \frac{183}{3M} \simeq 0.0142\%.$$

Thus, remaining with the over-quota large pool is still individually rational for a delegator, even though that pool is no longer operating in a fully reward-bearing region. Figure 5 visualizes the comparison.

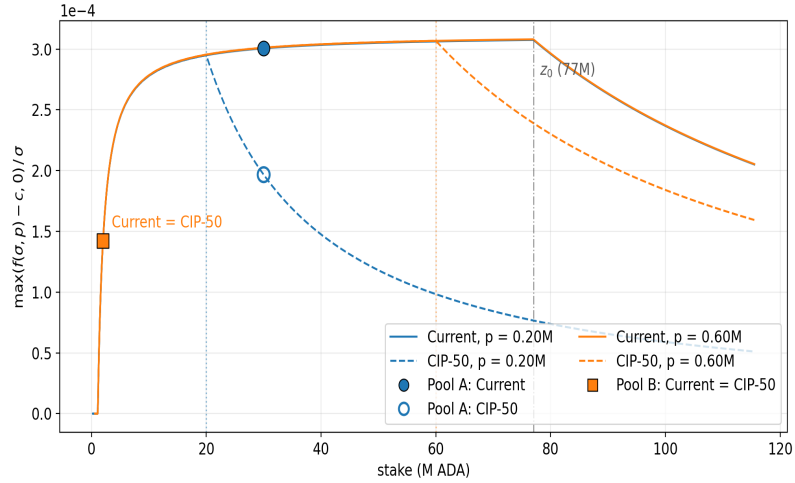


Figure 5: Two-pool illustration before and after CIP-50. After implementation, Pool A exceeds its quota whereas Pool B remains below it. Nevertheless, Pool A continues to offer a higher net reward per unit of stake

This example clarifies the distinction between feasibility and realized migration. The object $E(L)$ measures how much stake would need to move if all active stake were to remain in reward-bearing

¹²Use $\frac{f(\sigma_i, p_i; p_i L) - c}{\sigma_i} = \frac{1}{\sigma_i} \left\{ \frac{R}{(1+a_0)} \left[\frac{p_i}{T} + \frac{p_i}{T} \frac{a_0}{z_0} \left(\frac{p_i}{T} - \frac{p_i}{T} \left(\frac{z_0}{T} - \frac{p_i}{T} \right) \right) \right] - c \right\}$ to replicate the calculation.

positions. It does not imply that delegators will move one-for-one in that amount after the reform. Appendix C reports complementary heatmaps that illustrate the same point in a broader local state space.

Finding: Realized migration

A large $E(L)$ does not imply equally large realized migration. Some over-quota pools may continue to retain stake when their net returns remain more attractive than those of the available alternatives.

Notice that the previous argument about the limits in the realized migration relaxes the point about the transition pressure.

The analysis in this section provides a benchmark under the current configuration, not a full account of endogenous adaptation. In practice, operators may respond by increasing pledge, restructuring pool configurations, or reducing participation. These responses can relax the fixed-pledge transition problem, but they also shift attention toward broader policy questions about capital requirements, operator behavior, and other risks beyond feasibility. We turn to those next.

4 Policy-Relevant Risks Beyond Feasibility

The feasibility and transition problems discussed above are the central risks of a low value of L , but they do not exhaust the policy concerns raised by CIP-50. This section makes the broader risk map explicit by discussing additional implementation-related channels.

4.1 Risk taxonomy

Risk	Interpretation for policy
Aggregate reward-bearing shortfall	If $Q(L) < S$, the incumbent pool set cannot accommodate all currently active stake in reward-bearing positions under the current pledge distribution. This is a hard rejection criterion for the fixed-pledge benchmark.
Large reallocation requirement	Even if $Q(L) \geq S$, the current allocation may still be far from a fully reward-bearing post-reform allocation. $E(L)$ measures the minimum stake that must move away from currently over-quota pools.
Fragmented slack and user-coordination failure	Aggregate free space may exist only as many small pockets spread across pools. Discovering and coordinating on that space may be difficult for delegators.
Capital-based centralization / entry barrier	If L is low, some low-pledge pools can lose viability sharply, shifting competition from stake accumulation toward pledge capital. This can raise entry barriers and strengthen the position of operators with greater ability to post pledge.
Custodial-operator optimization	Large operators with internal stake-routing capacity may respond by optimizing pool structure, preserving inefficient structures, or reducing participation.
Delegation instability near quota	Under CIP-50, the reward schedule becomes kinked around the cap given by $p_i L$, making pool attractiveness more sensitive to stake changes near it.
Strategic crowding attacks	A rival may exploit that kink by pushing a target above its quota, worsening its return per unit of stake and trying to attract the resulting outflow.

Table 3: Policy-relevant risks under CIP-50.

The first three risks were discussed in the previous section. We therefore turn directly to the remaining ones.

4.2 Capital-based centralization and entry-barrier risk

CIP-50 is intended to weaken scale-based advantage by making pledge more relevant. But this creates a countervailing force: capital-based advantage. If L is low, then pools with weak pledge relative to their current stake lose reward-bearing capacity early. This may help well-pledged small and medium pools, but it may also exclude low-pledge operators from the competitive margin.

That risk matters for policy because decentralization is multi-dimensional. A reform can reduce one form of centralization while worsening another. In particular:

- it may reduce the attractiveness of large, weakly pledged pools;
- it may simultaneously make market access harder for operators with limited ability to post pledge;
- it may therefore shift competition from stake accumulation toward capital availability.

This paper does not claim that CIP-50 necessarily worsens decentralization overall. The point is narrower. A low L should not be marketed as an unambiguous decentralization improvement. It trades off scale discipline against stronger capital requirements.

Finding: Capital requirements and entry barriers

A low leverage value can shift competition away from stake and toward access to capital. This may raise entry barriers and strengthen the position of already well-capitalized operators.

4.3 Custodial-operator optimization

Another relevant channel concerns large custodial or exchange-like operators that control both a set of pools and a large stock of stake routed across them.¹³

To formalize this channel, let

$$W_g(L) := \Pi_g^*(L) - \Pi_g^{obs}(L),$$

where $\Pi_g^{obs}(L)$ is the total reward paid to operator group g under the observed pool configuration given certain L , and $\Pi_g^*(L)$ is the maximum reward that the same group could obtain after reallocating stake and pledge across its pools.

Under the current design, $W_g(\infty)$ can be positive for two broad reasons. First, some operators may simply not be running reward-maximizing pool structures. Second, they may be optimizing along other margins, such as operational simplicity, branding, legal segmentation, risk diversification, or a preference for keeping capital outside pledge. In either case, the observed configuration can pay less than a more reward-efficient one.

This wedge matters because rewards not paid to pools remain in reserves. A reward-inefficient pool structure therefore lowers current payouts and slows reserve depletion. CIP-50 can compress this wedge sharply. In particular, for a zero-pledge pool,

$$q_i(L) = \min\{z_0, p_i L\} = 0,$$

so a configuration that remains reward-bearing today ceases to do so under CIP-50.

A large operator facing this change has four broad responses: inject pledge, consolidate or restructure its pool set, keep the existing structure and accept lower paid rewards, or reduce participation. The policy implication is therefore ambiguous but important. CIP-50 may increase paid rewards by forcing more efficient pool structures, but it may also reduce active stake if some operators prefer not to adapt.

Finding: Custodial-operator response

CIP-50 can force large custodial operators to reconsider current reward-inefficient pool structures. Their response may range from more efficient pool structures, raising paid rewards and accelerating reserve depletion, to a reduction in participation.

4.3.1 Effects on reserve depletion

The full object $W_g(L)$ is not identified from the snapshot alone, because it would require the true operator partition and each group's feasible set for reallocating stake and pledge. We therefore compute a narrower proxy that is transparent and directly measurable.

¹³The March 3, 2026 snapshot shows why this matters. The data include 12 active eToro pools (matched by pool name/ticker) with zero declared pledge and about 0.472 B ADA of stake. More broadly, pools with zero declared pledge account for about 3.584 B ADA of active stake. These figures do not prove common ownership of all stake delegated to those pools, but they do show that economically meaningful amounts of stake currently sit in configurations that would be directly exposed under CIP-50.

The question is: Holding the observed stake allocation fixed, how much larger would the rewards of pools with current stake above the cap $p_i L$ be if they were given just enough additional pledge to bring that stake within the reward-bearing quota?

Define the normalized quantities

$$\bar{\sigma}_i := \frac{\sigma_i}{T}, \quad \bar{p}_i := \frac{p_i}{T}, \quad \bar{z}_0 := \frac{z_0}{T},$$

together with

$$\bar{\sigma}_i^* := \min\{\bar{\sigma}_i, \bar{z}_0\}, \quad \bar{p}_i^* := \min\{\bar{p}_i, \bar{z}_0\}.$$

Under the current design, pool i 's gross reward share of the epoch pool reward pot is

$$f_i^{obs} = \frac{1}{1 + a_0} \left[\bar{\sigma}_i^* + a_0 \bar{p}_i^* \frac{\bar{\sigma}_i^* - \bar{p}_i^* \frac{\bar{z}_0 - \bar{\sigma}_i^*}{\bar{z}_0}}{\bar{z}_0} \right].$$

For a leverage value L , define the set of affected pools by

$$A(L) := \{i \in P : L p_i < \sigma_i^*\}, \quad \sigma_i^* := \min\{\sigma_i, z_0\}.$$

For each $i \in A(L)$, the minimal pledge consistent with the current reward-relevant stake is

$$\tilde{p}_i(L) := \frac{\sigma_i^*}{L}, \quad \tilde{\tilde{p}}_i(L) := \frac{\bar{\sigma}_i^*}{L}.$$

Thus, the corresponding counterfactual reward share is

$$\tilde{f}_i(L) = \frac{1}{1 + a_0} \left[\bar{\sigma}_i^* + a_0 \tilde{\tilde{p}}_i(L) \frac{\bar{\sigma}_i^* - \tilde{\tilde{p}}_i(L) \frac{\bar{z}_0 - \bar{\sigma}_i^*}{\bar{z}_0}}{\bar{z}_0} \right].$$

Define

$$\delta(L) := \sum_{i \in A(L)} (\tilde{f}_i(L) - f_i^{obs}).$$

If R_{ep}^{pool} is the epoch reward pot allocated to pools after the treasury cut, the additional payout under this counterfactual is

$$\Delta^{epoch}(L) = R_{ep}^{pool} \delta(L), \quad \Delta^{year}(L) = 73 R_{ep}^{pool} \delta(L).$$

Equivalently, $\Delta^{epoch}(L)$ is the reduction in reserve carry-forward relative to the observed configuration.

In the March 3, 2026 snapshot, the calibration at the smallest admissible integer leverage $L = 141$ from the capacity calibration above gives

$$|A(141)| = 983, \quad \delta(141) \approx 3.57 \times 10^{-4}.$$

Thus, the stake-fixed non-optimization wedge is only about 0.0357% of the epoch pool reward pot. Equivalently, each 1 million ADA of epoch pool rewards corresponds to roughly 381 ADA of additional payout under the minimal-pledge counterfactual.

Some numerical scalings are useful. Using an illustrative value $R_{ep}^{pool} = 15.6$ million ADA,

$$\Delta^{epoch}(141) \approx 5,564 \text{ ADA}, \quad \Delta^{year}(141) \approx 0.406 \text{ M ADA}.$$

If we assume that the conditions observed in the snapshot persist over the coming years, the cumulative reserves retained after t years are simply

$$t \cdot \Delta^{year}(141).$$

Figure 6 plots this object over a 20-year. The endpoint magnitude is modest: about 4.74 million ADA under the illustrative 15.6 million ADA scaling.

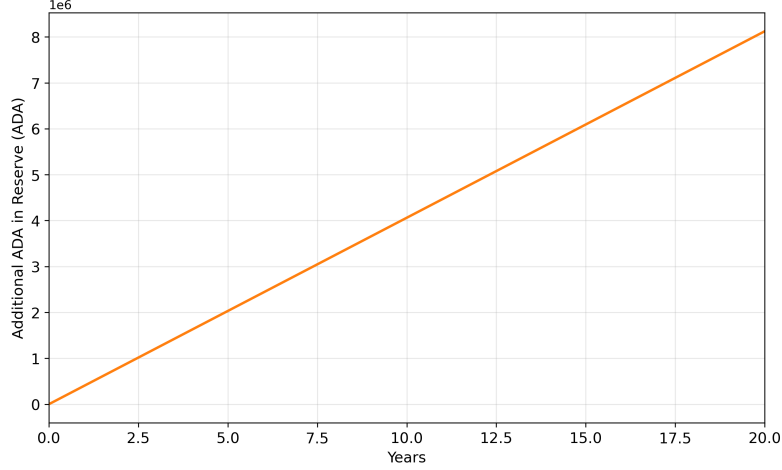


Figure 6: Cumulative reserve retained under continued non-optimization at $L = 141$.

Finding: Modest reserve effect

The increase in reserve depletion due to custodial-operator optimization does not appear to be quantitatively important.

4.4 Delegation instability and crowding attacks

As it was already mentioned, under the current design, many subsaturated pools face a relatively smooth reward environment. Under CIP-50, by contrast, each pool acquires a pledge-dependent reward kink at the quota $q_i(L) = \min\{z_0, p_i L\}$. This creates a sharper local deterioration in member returns once stake moves above the relevant quota.

The policy implication is that pool competitiveness can become more locally unstable around the kink:

- a pool slightly below quota may look attractive;
- a small additional inflow can push it above quota;
- once above quota, its reward per unit of stake worsens;
- this can trigger a reverse flow toward other pools.

This creates configurations in which crowding attacks may become strategically relevant. An attacker may temporarily add stake to a rival in order to push it beyond the reward kink and thereby induce delegation away from that rival.

This logic immediately restricts the set of plausible targets under the current design. Since the quota is the common saturation point z_0 , the natural targets are pools already close to saturation.

By contrast, a pool far below z_0 would require a much larger inflow of stake before its reward per unit begins to decline. The attacker, in turn, need not be large. What matters is not size per-se, but the ability to remain sufficiently attractive after the target has been crowded and delegators start to move. In that sense, under the current design the natural configuration is a smaller pool attacking a larger pool that is already close to z_0 .

CIP-50 preserves this basic strategic logic, but changes the structure of the problem in two ways. First, the common saturation point z_0 remains relevant, so the current-design channel does not disappear. A pool that is already close to z_0 may still be crowded in the same way as before. Second, however, CIP-50 introduces an additional reward kink for any pool whose pledge-based cap $p_i L$ lies below z_0 . As a consequence, there is no longer a single common kink. Instead, there is a family of pool-specific kinks, one for each pool whose pledge-based cap binds before z_0 .

This broadens the set of strategically exposed configurations. Under CIP-50, a pool may be moderate or even small in total stake and nevertheless be vulnerable if it operates close to its own pledge-based cap. In that sense, CIP-50 increases the number of potential attack vectors: it preserves the original crowding channel around z_0 , while also creating new opportunities around pool-specific caps.

It also changes the identity of the plausible attacker. Under the current design, the natural target would be typically a large near-saturated pool, and the attacker could be a smaller pool with enough spare capacity. Under CIP-50, one can also have the opposite configuration: a larger pool could attack a smaller or slightly smaller pool that is already close to its own pledge-based cap. We illustrate this last case in the following example. The purpose of the example is not to calibrate the empirical prevalence of such attacks, but to show that pool-specific quotas can create configurations in which strategic crowding becomes locally feasible.

Example 2. Assume $a_0 = 0.3$, $k = 500$, $R = 15.6M$ ADA, total supply $T = 38.5B$ ADA, $z_0 = T/k = 77,000,000$ ADA, and $L = 100$. Consider two pools under CIP-50. Pool 1 is pledge-constrained, so its effective reward-bearing quota is its own cap $p_1 L$, while Pool 2 has sufficiently high pledge so that its effective quota remains z_0 . We assume that only the latter pool is strategic, while the former is not.¹⁴

Suppose Pool 1 is already extremely close to its own cap, while Pool 2 is larger and better pledged. Margins are chosen so that delegators are initially indifferent.

Pool i	Pledge p_i	Relevant quota	Stake σ_i	Gross reward / stake r_i	Margin m_i
Pool 1 (target)	100,000	10,000,000	9,999,900	0.031181%	2.000%
Pool 2 (attacker)	1,000,000	75,000,000	60,000,000	0.0312899%	2.3412%

These margins equalize net returns:

$$(1 - m_1) \times r_1 \approx (1 - m_2) \times r_2 \approx 0.030557\%.$$

Now suppose that Pool-2 operator can move 1,000,000 ADA from Pool 2 to Pool 1.¹⁵ The new stakes become

¹⁴A more relaxed assumption would be to characterize Pool-1 as partially responsive rather than entirely non-strategic. This lower degree of responsiveness of a small pool could be justified by operational friction inherent in small-scale operations, like limited labor capacity.

¹⁵We acknowledge that a stake migration from one pool to another is not immediately effective, giving a time buffer to the attacked pool to react. However, since we assume that Pool-1 is unresponsive, the effect of these delays are irrelevant for the example.

<i>Pool</i>	<i>Pledge</i>	<i>Relevant quota</i>	<i>New stake</i>	<i>Comment</i>	<i>Gross reward / stake r_i</i>
<i>Pool 1</i>	100,000	10,000,000	10,999,900	<i>Above quota by 999,900</i>	0.0283465%
<i>Pool 2</i>	1,000,000	75,000,000	59,000,000	<i>Still below quota</i>	0.0312898%

Recomputing rewards from the CIP-50 formula gives

$$(1 - m_1) \times r_1 = 0.0277796\%, \quad (1 - m_2) \times r_2 = 0.0305572\%.$$

Pool 1 is pushed above its own quota, which lowers its reward per unit of stake and makes Pool 2 strictly more attractive to delegators. This creates an incentive for stake to migrate from Pool 1 to Pool 2. As a benchmark, suppose that 1,000,000 ADA migrates from Pool 1 to Pool 2, thereby restoring the two pools to their initial stake levels. Next, Pool 2 operator withdraws the temporary 1,000,000 ADA from Pool 1 and redelegates it to Pool 2. Final stakes then become

<i>Pool</i>	<i>Pledge</i>	<i>Final stake</i>	<i>Gross reward / stake r_i</i>
<i>Pool 1</i>	100,000	8,999,900	0.0311809%
<i>Pool 2</i>	1,000,000	61,000,000	0.0312899%

Notice that Pool 2 and Pool 1 are again with similar net returns:

$$(1 - m_1) \times r_1 = 0.0305572\%, \quad (1 - m_2) \times r_2 = 0.0305573\%,$$

and we could question the rationality of the attack. However, the attack may still be beneficial for the attacker in several dimensions:

- *The attack makes rewards per stake at Pool-1 very volatile while Pool-2 is always at a similar return ratio. Delegators may prefer those pools with less volatility since, otherwise, they would need to optimize their staking position frequently.*
- *We assumed that there was a migration from Pool-1 to Pool-2 of exactly the amount of the attack. However, migration would probably take time, and rewards are not equalized until the migration reaches 1M ADA .*

The practical relevance of this type of attack depends on timing, stake mobility, and the target's recapitalization ability. If the attacker adds $\Delta\sigma$ to the target, the target can neutralize this by adding only $\Delta p = \Delta\sigma/L$. Hence, a higher L makes defense cheaper relative to attack.

The main message of this section is that CIP-50 can create additional local vulnerability points around pool-specific quotas, especially when pools operate close to them and recapitalization or stake reallocation is slow, not that these kind of attacks will be a generalized behavior. Once pool-specific pledge-based caps become relevant, even comparatively small pools may become strategically exposed if they operate close to their own cap. This creates the possibility that a larger pool can strategically target a smaller pool through a crowding attack, something that does not arise naturally under the current design, where the reward vulnerability kink is the same for all pools at the saturation point z_0 .

Finding: Vulnerability points

Reward quotas create local vulnerability points at the kinks. Under the current design, this is concentrated at the common saturation point z_0 . Under CIP-50, it becomes more widespread, since each pool with $p_i L < z_0$ acquires its own vulnerability point at $p_i L$.

Finding: Strategic exploitation

These points may become strategically exploitable when defense is delayed and pledge mobility is limited.

Finding: Higher delegator monitoring requirement

Delegators may need to monitor both their own pool and the broader pool landscape more frequently. This may worsen the user experience.

5 Conclusion

CIP-50 is motivated by real concerns. Under the current reward design, pledge appears too weakly connected to delegator-facing returns for most pools, while stake size and fixed-cost dilution seem to play a larger role in shaping competitive outcomes. This may favor Sybil-type behavior and, over time, contribute to greater centralization. The proposal aims to mitigate these issues.

However, CIP-50 does not merely change the marginal role of pledge in the reward formula. It changes the effective reward-bearing quota by replacing a one-cap design with a two-cap design, in which each pool may face not only the common saturation point z_0 , but also its own pledge-based cap $p_i L$. This makes the choice of L central since it might induce some pools to increase pledge, but also it may create other market and behavioral frictions.

The first result of the paper is therefore a simple admissibility criterion: any proposed value of L must first satisfy the reward-bearing feasibility condition $Q(L) \geq S$. If it does not, the incumbent pool set cannot accommodate all currently active stake in reward-bearing positions under the current pledge distribution. In practice, that pressure may be absorbed through some combination of added pledge, restructuring, stake reallocation, or reduced participation.

The second result is that feasibility is not enough. Even when L is just above the threshold, transition pressure can remain severe. Large amounts of stake may still need to move away from pools that are currently above quota in order to reach a fully reward-bearing allocation under the fixed-pledge benchmark. Hence, passing the aggregate feasibility test does not imply a smooth transition. It only means that a fully reward-bearing allocation exists in principle.

The transition problem is also two-dimensional. A low value of L can generate pressure not only through stake reallocation, but also through the need for additional pledge. In the current snapshot, the recapitalization margin appears much smaller than the pure reallocation margin, suggesting that endogenous operator responses may materially affect how difficult the transition proves in practice.

The paper also shows that the transition problem is not purely mechanical. Delegators respond to expected net returns, not to feasibility conditions directly, so some stake may remain in over-quota pools if those pools still offer relatively attractive returns after fixed costs. This creates a gap between the accounting reallocation requirement and realized migration.

Beyond feasibility and reallocation, the reform creates additional policy-relevant risks. A low L strengthens capital requirements and may shift competitive advantage from scale toward capital, with ambiguous implications for decentralization. At the same time, the new pool-specific quotas create more local vulnerability points in the reward schedule, which can make delegation more unstable

around the kinks and open the door to potential exploitation of these vulnerabilities.

The broader conclusion is therefore that CIP-50 should be evaluated as a broader transition problem rather than as a simple change in incentives.

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A Current State: Descriptive Statistics

This section summarizes the March 3, 2026 snapshot used in the paper (see data set description in Appendix B for more details). Unless otherwise stated, all statistics refer to *active* pools, defined as pools with strictly positive active stake.

A.1 Sample construction and aggregate quantities

Metric	Value
Pools in source	2941
Excluded inactive pools (<i>active_stake</i> = 0)	224
Active pools used in the analysis	2,717

Table 4: Sample construction.

Metric	Value
Total active stake	21568.605M ADA
Total declared pledge	2577.398M ADA
Average active stake per pool	7.938M ADA
Average declared pledge per pool	0.949M ADA

Table 5: Aggregate and average quantities across active pools. Declared pledge is certificate pledge (`pool_update.active.pledge`).

A.2 Saturation, zero pledge, and self-staked pools

Category	Pools	% of pools	Stake	% of active stake
Saturation ≥ 1	7	0.26%	0.7B	3.25%
Saturation ≥ 0.98	27	0.99%	2.2B	10.28%
Declared pledge = 0	225	8.28%	3.6B	16.62%
Active stake = active pledge	329	12.11%	0.3B	1.41%

Table 6: Selected pool categories in the active sample.

Two patterns are worth highlighting. First, very few pools are close to full saturation. Second, zero-declared-pledge pools are not rare: they account for 225 active pools and 16.62% of all active stake. By contrast, pools with active stake equal to active pledge are more numerous but collectively small in stake terms.

A.3 Stake distribution across all active pools

Across all 2,717 active pools (that is, pools with *active_stake* > 0), stake is also concentrated in higher stake bins.

Stake range (M ADA)	Pools	% pools	Aggregate stake	% of total stake
[0,10)	2205	81.16%	1780.2M	8.25%
[10,20)	146	5.37%	2080.1M	9.64%
[20,30)	65	2.39%	1593.2M	7.39%
[30,40)	92	3.39%	3154.2M	14.62%
[40,50)	44	1.62%	1973.5M	9.15%
[50,60)	55	2.02%	3019.1M	14.00%
[60,70)	31	1.14%	2007.7M	9.31%
[70,80]	72	2.65%	5257.5M	24.38%
≥ 80	7	0.26%	703.0M	3.26%

Table 7: Stake distribution among all active pools (*active_stake* > 0).

A.4 Pools with zero declared pledge

There are 225 active pools with *declared_pledge* = 0. Their stake distribution is highly concentrated: most such pools are small, but a relatively small number of large zero-declared-pledge pools account for most of the subgroup’s stake.

Stake range (M ADA)	Pools	% of subgroup	Aggregate stake	% of subgroup stake	% of total stake
[0,10)	156	69.33%	56.9M	1.59%	0.26%
[10,20)	9	4.00%	136.9M	3.82%	0.63%
[20,30)	3	1.33%	76.6M	2.14%	0.36%
[30,40)	11	4.89%	383.4M	10.70%	1.78%
[40,50)	3	1.33%	135.6M	3.78%	0.63%
[50,60)	14	6.22%	746.6M	20.83%	3.46%
[60,70)	5	2.22%	313.4M	8.74%	1.45%
[70,80]	23	10.22%	1614.6M	45.05%	7.49%
≥ 80	1	0.44%	120.4M	3.36%	0.56%

Table 8: Stake distribution among active pools with zero declared pledge.

Fee choices in this subgroup.

Margin group	Count	% of subgroup
0	50	22.22%
(0,0.1]	132	58.67%
(0.1,0.2]	3	1.33%
(0.2,0.3]	2	0.89%
(0.9,1)	10	4.44%
=1	28	12.44%

Table 9: Margin distribution among active pools with zero declared pledge. Empty intermediate bins omitted.

Fixed cost group (ADA)	Count	% of subgroup
(0,340)	18	8.00%
=340	185	82.22%
(340,500)	15	6.67%
=500	3	1.33%
≥ 500	4	1.78%

Table 10: Fixed cost distribution among active pools with zero declared pledge.

Most zero-declared-pledge pools use low margins and the standard fixed cost of 340 ADA. This suggests that zero declared pledge is not mechanically associated with unusually aggressive fee choices.

A.5 Pools with active stake equal to active pledge

A second relevant subgroup consists of pools with *active_stake* = *active_pledge* (live pledge). These are pools with no delegated stake at the snapshot date. A natural conjecture is that such pools may be intentionally discouraging delegation through high fees, but the fee distributions provide limited support for that interpretation.

Margin group	Count	% of subgroup
0	49	14.89%
(0,0.1]	223	67.78%
(0.1,0.2]	22	6.69%
(0.2,0.3]	1	0.30%
(0.8,0.9]	1	0.30%
(0.9,1)	3	0.91%
=1	30	9.12%

Table 11: Margin distribution among pools with active stake equal to active pledge. Empty intermediate bins omitted.

Fixed cost group (ADA)	Count	% of subgroup
(0,340)	31	9.42%
=340	218	66.26%
(340,500)	72	21.88%
=500	7	2.13%
(500,1000)	1	0.30%

Table 12: Fixed cost distribution among pools with active stake equal to active pledge.

Most of these pools also cluster at low margins and standard fixed costs. Hence the absence of delegated stake in this subgroup should not be interpreted solely as a fee-setting phenomenon. Other explanations, such as recency, limited visibility, or weak demand from delegators, are also plausible.

B Delegator choice: panel evidence.

This appendix provides a more detailed empirical look at what appears to drive delegator choice across pools. The main result is simple: **current stake** is positively associated with delegator growth, while **declared pledge** is not a robust predictor.

Main finding.

- Evidence suggests a robust positive relationship between lagged stake and delegator growth.
- We do not find robust evidence of a positive relationship between declared pledge and delegator growth.

Data Set. The empirical analysis uses a pool-level dataset for Cardano stake pools based on a snapshot dated March 3, 2026 (i.e., during epoch 616). The raw file contains 2,941 pools and 169 variables. It combines current pool-level information—including active stake, live stake, pledge, delegators, margin, fixed cost, and ROA measures—with the last 8 epoch-level histories on stake, pledge, delegators, rewards, and block production. For the main calculations, the sample is restricted to pools with strictly positive active stake, which leaves 2,717 active pools.

Model. To evaluate whether declared pledge is relevant for delegator growth with the available data, we estimate correlated random effects (Mundlak) specifications.

We estimate correlated random effects (Mundlak) panel equations:

$$\Delta \log(1 + delegators_{it}) = \beta_0 + \beta_1 \log(1 + declared_pledge_i) + \beta_2 X_{i,t-1} + \beta_3 \bar{X}_i + \gamma_t + u_i + \varepsilon_{it}.$$

where $\Delta \log(1 + delegators_{it}) = \log(1 + delegators_{it}) - \log(1 + delegators_{i,t-1})$, $X_{i,t-1}$ includes lagged time-varying controls, $\bar{X}_i$ are pool means (Mundlak terms), and γ_t are epoch fixed effects. Standard errors are clustered by pool.

Key terms.

- **Declared β :** estimated association between declared pledge and delegator growth, conditional on controls.
- **Declared p -value:** statistical evidence against zero effect; lower values indicate stronger evidence.
- **Negative declared β :** higher declared pledge is associated with lower delegator growth in that specification (association, not causality).
- **Persistence:** delegation dynamics are inertial; current outcomes depend on previous delegator levels.

CRE1. Baseline CRE with lagged luck and lagged stake.

Variable	Beta	Cluster SE (pool)	p-value
$\log(1 + \text{declared pledge})$	-0.000073	0.000042	0.0842
lag1 luck	-0.000458	0.000575	0.4262
lag1 log stake	0.014800	0.003646	0.0000
pool mean lag1 luck	0.000923	0.000958	0.3355

Variable	Beta	Cluster SE (pool)	p-value
pool mean lag1 log stake	-0.014609	0.003636	0.0001
epoch 610	0.002791	0.001747	0.1102
epoch 613	-0.000155	0.000994	0.8760
epoch 614	0.001144	0.000829	0.1676
epoch 615	-0.000067	0.001228	0.9567

CRE2 Adds lagged delegators to capture persistence in delegation dynamics (inertia/base effects).

Variable	Beta	Cluster SE (pool)	p-value
log(1 + declared pledge)	-0.000105	0.000046	0.0237
lag1 luck	-0.000144	0.000301	0.6327
lag1 log stake	0.022816	0.004688	0.0000
lag1 log delegators	-0.633104	0.140292	0.0000
pool mean lag1 luck	0.001137	0.001047	0.2776
pool mean lag1 log stake	-0.022444	0.004649	0.0000
pool mean lag1 log delegators	0.632217	0.140296	0.0000
epoch 610	0.001545	0.000887	0.0818
epoch 613	0.001490	0.001066	0.1624
epoch 614	0.003221	0.001118	0.0040
epoch 615	0.003350	0.001449	0.0208

CRE3 Adds lagged active pledge to separate declared and active-pledge channels and test whether declared pledge carries independent information.

Variable	Beta	Cluster SE (pool)	p-value
log(1 + declared pledge)	-0.000043	0.000044	0.3279
lag1 luck	-0.000150	0.000302	0.6198
lag1 log stake	0.022795	0.004682	0.0000
lag1 log delegators	-0.633387	0.140348	0.0000
lag1 log active pledge	0.001843	0.001574	0.2418
pool mean lag1 luck	0.001353	0.001074	0.2081
pool mean lag1 log stake	-0.022287	0.004638	0.0000
pool mean lag1 log delegators	0.632484	0.140347	0.0000
pool mean lag1 log active pledge	-0.002182	0.001656	0.1877
epoch 610	0.001241	0.000804	0.1228
epoch 613	0.001189	0.001005	0.2370
epoch 614	0.002958	0.001056	0.0051
epoch 615	0.003087	0.001389	0.0263

C Local incentive maps for delegator choice

To complement the two-pool example in the main text, this appendix reports local incentive maps for an incoming delegator choosing between a Large Pool and a Small Pool.

Figures 7 and 8 plot the difference in net reward per unit of stake between the Large Pool and the Small Pool for different combinations of Small-Pool pledge and post-delegation stake. Positive values indicate regions in which the Large Pool offers the higher net return per unit of stake; negative values indicate regions in which the Small Pool is more attractive. The dashed curve marks the indifference frontier.

The first figure corresponds to the current reward design. It shows that scale can remain a strong advantage over a broad region, even when the smaller pool has more pledge. The second figure repeats the exercise under CIP-50. The comparison becomes more pledge-sensitive, and the better-pledged Small Pool gains ground over a wider set of states. At the same time, the maps also show that CIP-50 does not imply one-for-one migration away from all over-quota large pools: there remain regions in which the Large Pool continues to offer the higher net return per unit of stake.

These figures should therefore be read as local incentive maps. Their role is not to characterize the full equilibrium allocation, but to illustrate why transition pressure and realized migration need not coincide.

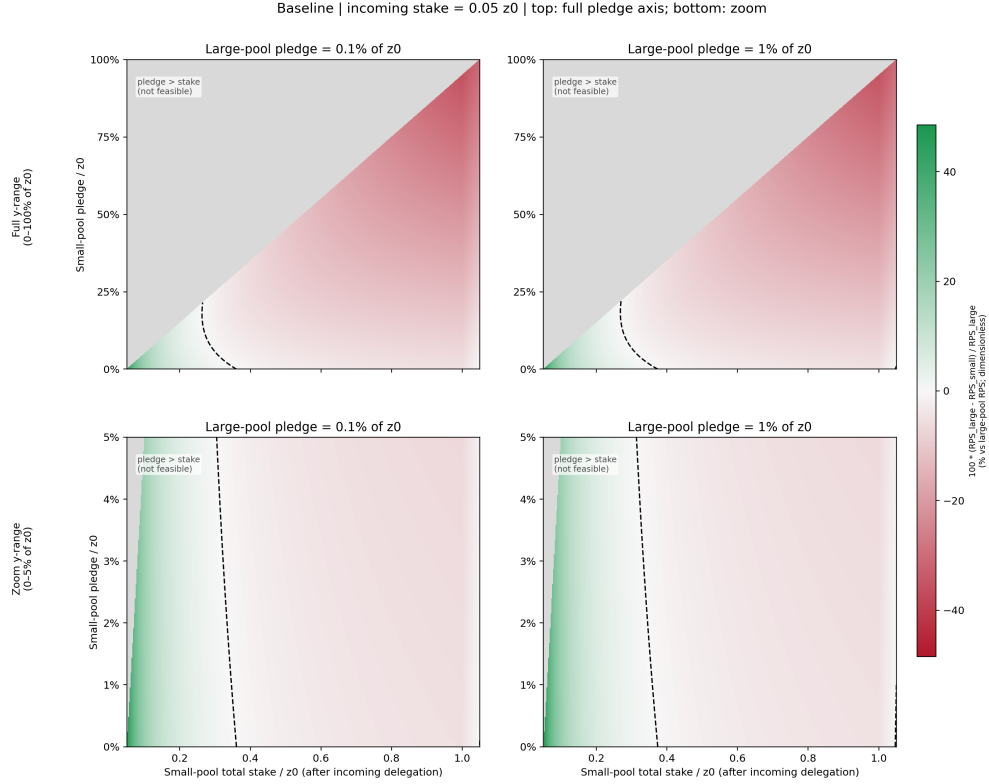


Figure 7: Delegator-choice heatmaps under the current reward design. Each panel compares the net reward per unit of stake offered by a Large Pool (with stake z_0) and a Small Pool after an incoming delegation. Positive values favor the Large Pool; negative values favor the Small Pool. The dashed curve marks the indifference frontier. At the bottom the plots are zoomed.

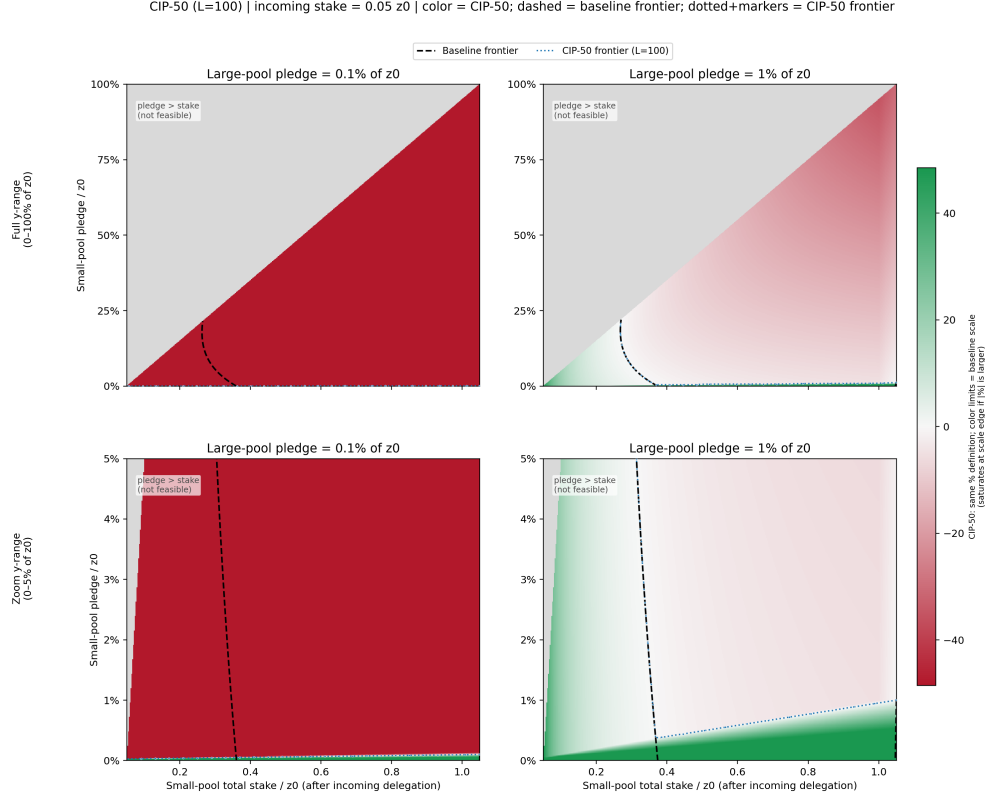


Figure 8: Delegator-choice heatmaps under CIP-50 with $L = 100$. For $L = 100$, $p \cdot L > z_0$ already for $p = 1\%$. Hence, at the bottom the plots are zoomed. Positive values favor the Large Pool; negative values favor the Small Pool. The dashed curve marks the indifference frontier.

D Lists

We list the main operators with zero-pledge pools and the main operators with pools in the $(0, 1000]$ pledge range. These figures should be interpreted with caution, since some large operators run pools under different names that do not clearly reveal a common identity, making operator-level identification difficult.

D.1 Main operators with zero-declared pledge pools

Operator	# pools	Tot. Stake (M ADA)	Avg. Stake (M ADA)
Unidentified	110	1,556.415	14.149
etoro.com	12	471.870	39.322
stakebowl.io	7	140.045	20.006
Kind Orange Heisenberg	1	70.537	70.537
Upbeat Beige Euler	1	70.335	70.335

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Operator	# pools	Tot. Stake (M ADA)	Avg. Stake (M ADA)
Wacky Gold Maxwell	1	70.325	70.325
Thin Blue Galileo	1	70.199	70.199
Admiring Red Hertz	1	70.189	70.189
Fervent Vermilion	1	70.078	70.078
Galileo			
Lucky Azure Gauss	1	70.050	70.050
Ubiquitous Teal Aslan	1	70.048	70.048
Tidy Lilac Kepler	1	70.039	70.039
Tall Zaffre Darwin	1	70.028	70.028
Relaxed Mustard Hop- per	1	70.024	70.024
Fancy Mango Dirac	1	70.021	70.021
Fancy Burgundy	1	70.019	70.019
Archimedes			
infstones.com	1	61.619	61.619
Keen Lemon Fermi	1	50.263	50.263
Elegant Gold Heisen- berg	1	35.448	35.448
Thoughtful Iris Albat- tani	1	35.040	35.040
Curly Viridian Moore	1	35.024	35.024
Responsible Rose	1	35.014	35.014
Coanda			
Futuristic Cadet	1	35.012	35.012
Swartz			
Grandiose Charcoal	1	35.012	35.012
Nobel			
Gleaming Cyan	1	35.011	35.011
Davinci			
krypto-knight.us	1	34.481	34.481
QQQ	1	25.207	25.207
theswim.net	2	16.248	8.124
DEVFOG	1	12.222	12.222
Ragdoll	1	10.040	10.040
p2p.org	1	7.276	7.276
londonpool.co.uk	2	6.911	3.456
.	1	5.939	5.939
BRADA POOL	1	5.666	5.666
socious.io	2	5.465	2.732
weqntstakepool.nl	1	4.303	4.303
nuvoladigital.io	1	2.859	2.859
sebastiengllmt.com	1	2.044	2.044
milkomeda.com	1	1.942	1.942

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Operator	# pools	Tot. Stake (M ADA)	Avg. Stake (M ADA)
cheapstaking.com	1	1.102	1.102
nespool.com	1	0.753	0.753
ALPEN Pool for De- velopment	1	0.575	0.575
bitboycrypto.com	1	0.528	0.528
themoneypool.uk	1	0.498	0.498
logicalmechanism.io	1	0.428	0.428
ada.zone	1	0.376	0.376
cardanolibrary.net	1	0.292	0.292
onyxstakepool.com	1	0.261	0.261
staking-cardano.com	1	0.231	0.231
coinbase.com	1	0.199	0.199
poolman.io	1	0.166	0.166
lgbtqpool.com	1	0.142	0.142
pools.cardano.icodeit.io	1	0.107	0.107
libraryofbabel.info	1	0.101	0.101
responsible-staking.at	1	0.072	0.072
ada-thailand.com	1	0.065	0.065
sheinixstaking.github.io	1	0.035	0.035
hudo.online	1	0.024	0.024
ilukapool.com	1	0.011	0.011
skyw.tech	1	0.011	0.011
ace.tokenodynamics.net	1	0.009	0.009
staked.us	1	0.009	0.009
aivs-	1	0.008	0.008
adapool.datamaven.pro			
cardano- education.com	1	0.007	0.007
elevate-staking.com	1	0.007	0.007
islandpool.github.io	1	0.006	0.006
pipemind.com	1	0.006	0.006
bitpullstake.com	1	0.006	0.006
decentralizedreality.io	1	0.005	0.005
neoply.io	1	0.003	0.003
stakinghouse.com	1	0.002	0.002
laobai.io	1	0.002	0.002
adastrium.io	1	0.002	0.002
derpbirds.io	1	0.002	0.002
-	1	0.001	0.001
sireto.com	1	0.001	0.001
kubecoin.org	1	0.001	0.001
FKAIPool	1	0.001	0.001
Proud Staking	1	0.001	0.001

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Operator	# pools	Tot. Stake (M ADA)	Avg. Stake (M ADA)
arubastakepool.com	1	0.000	0.000
sonicstaking.org	1	0.000	0.000
ironstaking.space	1	0.000	0.000
cardanospo.computeportal.net	1	0.000	0.000
ctrl.xyz	1	0.000	0.000
Rampant Violet	1	0.000	0.000
Hawking			
atherchain.io	1	0.000	0.000
Optimistic Olive Hy- patia	1	0.000	0.000
tidepools.io	1	0.000	0.000
brazilpool.com	1	0.000	0.000
moongangx.com	1	0.000	0.000
haxmode.io	1	0.000	0.000
cactus- stakepool.github.io	1	0.000	0.000
foo.com	1	0.000	0.000
raw.githubusercontent.com	1	0.000	0.000
ANON LowFee	1	0.000	0.000
thescenepool.com	1	0.000	0.000

D.2 Main operators with $(0, 1000]$ -declared pledge

Operator	# pools	Tot. Stake (M ADA)	Avg. Stake (M ADA)	Tot. pledge (ADA)	Avg. pledge (ADA)
kiln.fi	9	636.424	70.714	900.00	100.00
everstake.one	15	566.973	37.798	11,040.00	736.00
binance.com	36	424.065	11.780	72.00	2.00
Unidentified	340	395.785	1.164	104,322.34	306.83
figment.io	22	361.241	16.420	2.00	0.09
nu.fi	17	301.668	17.745	17,000.00	1,000.00
ledger.com	4	269.742	67.435	8.00	2.00
blockdaemon.com	6	244.580	40.763	1,200.00	200.00
emurgo.io	7	238.699	34.100	3,500.00	500.00
(Crypto Capital Ven- ture)	5	177.724	35.545	2,500.00	500.00
garden-pool.com	5	170.390	34.078	5,000.00	1,000.00
bitvavo.com	4	152.594	38.149	8.00	2.00
trustwallet.com	2	116.061	58.031	200.00	100.00
autostake.com	4	88.905	22.226	400.00	100.00
SASA Stake (same metadata description)	2	74.534	37.267	160.00	80.00

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Operator	# pools	Tot. Stake (M ADA)	Avg. Stake (M ADA)	Tot. pledge (ADA)	Avg. pledge (ADA)
ada.thenorthminer.com	1	65.406	65.406	1,000.00	1,000.00
tangem.com	1	61.438	61.438	500.00	500.00
adapool.biglazycat.com	2	61.019	30.509	2,000.00	1,000.00
stakepoolservice.com	3	56.862	18.954	3,000.00	1,000.00
cardano-tera.com	3	52.985	17.662	900.00	300.00
dappcentral.net	2	49.589	24.794	2,000.00	1,000.00
wavepool.digital	4	48.999	12.250	20.00	5.00
stakeshark.io	3	47.553	15.851	3,000.00	1,000.00
xray.app	8	45.670	5.709	8,000.00	1,000.00
planetarium39	1	45.333	45.333	500.00	500.00
sunnyada.io	2	45.318	22.659	2,000.00	1,000.00
forms.gle	3	42.308	14.103	300.00	100.00
luganodes.com	1	41.336	41.336	10.00	10.00
qzsbu	1	36.719	36.719	2.00	2.00
adapools.org	3	35.699	11.900	15.00	5.00
coffeepool.jp	2	34.016	17.008	202.00	101.00
ibexpool.com	2	31.666	15.833	8.00	4.00
bitpoint.co.jp	1	28.263	28.263	5.00	5.00
cardano.fans	1	27.671	27.671	1,000.00	1,000.00
bitcoinsuisse.com	1	27.073	27.073	100.00	100.00
coin-tax.net	1	26.522	26.522	100.00	100.00
Hisa~iOkinawa	1	25.466	25.466	100.00	100.00
muen-project.com	2	25.463	12.732	20.00	10.00
U0001	1	21.882	21.882	50.00	50.00
Well-being Financial Lounge	1	20.689	20.689	399.00	399.00
NVE Tech StakePool	1	19.550	19.550	100.00	100.00
AID POOL	1	19.161	19.161	80.00	80.00
MIMICPool	1	18.788	18.788	200.00	200.00
p2p.org	2	16.190	8.095	2,000.00	1,000.00
bitapool.wixsite.com	1	14.170	14.170	10.00	10.00
validator.cex.io	1	13.470	13.470	1,000.00	1,000.00
cardanistas.io	1	12.972	12.972	1,000.00	1,000.00
growpools.io	1	12.937	12.937	555.00	555.00
ada.ibexpool.com	1	12.602	12.602	5.00	5.00
MYUJK Pool	1	12.118	12.118	1,000.00	1,000.00
piertwo.com	2	12.014	6.007	200.00	100.00
ellipal.com	1	11.790	11.790	1,000.00	1,000.00
zjavadoc.github.io	1	11.548	11.548	30.00	30.00
osaka-pool.com	1	11.015	11.015	1,000.00	1,000.00
coinex.com	1	10.851	10.851	2.00	2.00
yeatanotherstakepool.org	1	10.756	10.756	15.00	15.00
TreFi_Capital	1	10.562	10.562	10.00	10.00

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Operator	# pools	Tot. Stake (M ADA)	Avg. Stake (M ADA)	Tot. pledge (ADA)	Avg. pledge (ADA)
infiniastakepool.com	3	9.899	3.300	504.00	168.00
cardano.ipclub29.com	2	9.220	4.610	200.00	100.00
Kaze-Wa-Minamikara	1	8.748	8.748	1,000.00	1,000.00
tinyurl.com	2	8.624	4.312	200.00	100.00
cardano-thailand.com	1	8.613	8.613	100.00	100.00
KOG 01	1	8.400	8.400	80.00	80.00
spectrum.fi	1	8.308	8.308	400.00	400.00
thc-room335.com	2	8.007	4.004	2,000.00	1,000.00
yoroi-wallet.com	1	7.587	7.587	500.00	500.00
tap-ada.github.io	1	7.429	7.429	5.00	5.00
sqqq.work	1	7.262	7.262	100.00	100.00
ayame3.com	1	7.188	7.188	1.00	1.00
SMAN8	1	7.154	7.154	168.00	168.00
New One Stake Pool	1	7.073	7.073	100.00	100.00
kysenpool.io	2	6.898	3.449	200.00	100.00
lybiasoft.com	1	6.685	6.685	1,000.00	1,000.00
VoltPool	1	6.516	6.516	1,000.00	1,000.00
iagon.com	3	6.512	2.171	120.00	40.00
pegasuspool.info	1	6.359	6.359	500.00	500.00
RIPI Stake Pool	1	5.594	5.594	1.00	1.00
WSKS	1	5.417	5.417	100.00	100.00
ATARASHIIMONO Stake Pool	1	5.225	5.225	1,000.00	1,000.00
cryptoblocks.pro	1	5.192	5.192	1,000.00	1,000.00
mimosa	1	5.091	5.091	100.00	100.00
createplanets.com	1	5.051	5.051	1.00	1.00
bcw.group	1	5.028	5.028	1.00	1.00
Diamond	1	4.987	4.987	1,000.00	1,000.00
PlanningT	1	4.841	4.841	1,000.00	1,000.00
miyabi-kyoto.org	1	4.812	4.812	1,000.00	1,000.00
wingriders.com	1	4.196	4.196	100.00	100.00
Sagittarius Pool	1	4.049	4.049	1,000.00	1,000.00
HIT Pool	1	3.846	3.846	500.00	500.00
SPP Stake Pool	1	3.834	3.834	100.00	100.00
davidlikescrypto.com	1	3.589	3.589	100.00	100.00
cswap.info	2	3.501	1.751	2,000.00	1,000.00
cardapool.com	1	3.464	3.464	1.00	1.00
white.topopool.com	1	3.439	3.439	50.00	50.00
lidonation.com	1	3.354	3.354	1,000.00	1,000.00
frogs- mining.gitbook.io	1	3.334	3.334	100.00	100.00
OKAYAMA	1	3.316	3.316	1,000.00	1,000.00
no-tax.biz	2	3.304	1.652	1,500.00	750.00

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Operator	# pools	Tot. Stake (M ADA)	Avg. Stake (M ADA)	Tot. pledge (ADA)	Avg. pledge (ADA)
skylightpool.com	1	3.163	3.163	10.00	10.00
adabreathes.com	1	3.002	3.002	500.00	500.00
0pctpools.com	1	3.000	3.000	10.00	10.00
dconecrypto.finance	1	2.827	2.827	100.00	100.00
koalako.io	1	2.782	2.782	500.00	500.00
worldmobile.club	1	2.718	2.718	1,000.00	1,000.00
jorgepool.com	1	2.699	2.699	1,000.00	1,000.00
ultra-labs.io	1	2.667	2.667	1,000.00	1,000.00
Kool Stake Pool	1	2.577	2.577	1,000.00	1,000.00
usenobi.com	1	2.568	2.568	3.00	3.00
stakeforcake.com	1	2.501	2.501	500.00	500.00
outerspace.money	1	2.428	2.428	100.00	100.00
SEBI	1	2.357	2.357	1,000.00	1,000.00
cardanoaldia.com	1	2.344	2.344	1,000.00	1,000.00
adamoney.net	1	2.331	2.331	1,000.00	1,000.00
kuris.org	1	2.311	2.311	100.00	100.00
HRDIB	1	2.284	2.284	500.00	500.00
adoptstakepool.com	1	2.142	2.142	250.00	250.00
cardistack.cryptoconnect.uk	1	2.092	2.092	10.00	10.00
tkgems.com	1	1.962	1.962	1,000.00	1,000.00
adavn.com	1	1.850	1.850	1,000.00	1,000.00
junostakepool.com	1	1.765	1.765	1,000.00	1,000.00
pool.bio	1	1.650	1.650	1,000.00	1,000.00
onlyblocks.eu	1	1.613	1.613	1,000.00	1,000.00
staketrek.io	1	1.514	1.514	1,000.00	1,000.00
Bitcã Stake Pool	1	1.512	1.512	1.00	1.00
cardano-aden.com	1	1.429	1.429	100.00	100.00
adastarnode.com	1	1.396	1.396	1,000.00	1,000.00
novapool.io	1	1.384	1.384	1,000.00	1,000.00
cardanolink.net	1	1.355	1.355	5.00	5.00
funontheride.com	1	1.350	1.350	1,000.00	1,000.00
houseoftitans.io	1	1.304	1.304	1,000.00	1,000.00
nizi-stake-pool.com	1	1.247	1.247	100.00	100.00
kurdopia.store	1	1.169	1.169	500.00	500.00
eldenpool.com	1	1.146	1.146	1,000.00	1,000.00
Nirvana	1	1.050	1.050	500.00	500.00
zeropercentstaking.com	1	1.017	1.017	1,000.00	1,000.00
Virtual Flame Studios	1	1.016	1.016	500.00	500.00
Others	441	31.218	0.071	153,136.75	347.25